

Practical-05:

```
GNU nano 8.7.1 producer_consumer.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

#define N 100000

int main()
{
    int pipefd[2];

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid > 0)
    {
        // Parent process - Producer
        close(pipefd[0]);

        int buffer[1024];
        int count = 0;
        long long total_bytes = 0;

        struct timespec start, end;
        clock_gettime(CLOCK_MONOTONIC, &start);

        for (int i = 1; i <= N; i++)
        {
            buffer[count++] = i;

            if (count == 1024 || i == N)
            {
                ssize_t bytes = write(pipefd[1],
                                      buffer,
                                      count * sizeof(int));

                if (bytes == -1)
                {
                    perror("write");
                    close(pipefd[1]);
                    wait(NULL);
                    return 1;
                }

                total_bytes += bytes;
                count = 0;
            }
        }

        close(pipefd[1]);

        wait(NULL);

        clock_gettime(CLOCK_MONOTONIC, &end);

        double elapsed =
            (end.tv_sec - start.tv_sec) +
            (end.tv_nsec - start.tv_nsec) / 1e9;

        double throughput =
            (total_bytes / (1024.0 * 1024.0)) / elapsed;

        printf("\nProducer: Generated %d integers.\n", N);
        printf("Producer: Sent %.2f KB through pipe.\n",
              total_bytes / 1024.0);
        printf("Communication Time: %.6f seconds\n", elapsed);
        printf("Communication Throughput: %.2f MB/s\n",
              throughput);
    }
}
```

```

        throughpipe,
    }
    else
    {
        // Child process - Consumer
        close(pipefd[1]);

        int buffer[1024];
        ssize_t bytes;
        long long sum = 0;
        long long count = 0;

        while ((bytes = read(pipefd[0],
                             buffer,
                             sizeof(buffer))) > 0)
        {
            int numbers = bytes / sizeof(int);

            for (int i = 0; i < numbers; i++)
            {
                sum += buffer[i];
                count++;
            }
        }

        close(pipefd[0]);

        printf("Consumer: Received %lld integers.\n", count);
        printf("Consumer: Sum of received data = %lld\n", sum);
        exit(0);
    }

    return 0;

```

Output:

```

uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./producer
Consumer: Received 100000 integers.
Consumer: Sum of received data = 5000050000

Producer: Generated 100000 integers.
Producer: Sent 390.62 KB through pipe.
Communication Time: 0.001226 seconds
Communication Throughput: 311.11 MB/s

```

2nd:

```
GNU nano 8.7.1 ls_grep_pipe.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    pid_t child1 = fork();

    if (child1 < 0)
    {
        perror("fork");
        return 1;
    }

    if (child1 == 0)
    {
        // First child: execute ls -l
        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[1]);

        execlp("ls", "ls", "-l", (char *)NULL);

        perror("execlp ls");
        exit(1);
    }

    pid_t child2 = fork();

    if (child2 < 0)
```

```
    {
        perror("fork");
        return 1;
    }

    if (child2 == 0)
    {
        // Second child: execute grep ".c"
        close(pipefd[1]);

        dup2(pipefd[0], STDIN_FILENO);
        close(pipefd[0]);

        execlp("grep", "grep", ".c", (char *)NULL);

        perror("execlp grep");
        exit(1);
    }

    // Parent process
    close(pipefd[0]);
    close(pipefd[1]);

    waitpid(child1, NULL, 0);
    waitpid(child2, NULL, 0);

    printf("\nCommand executed successfully: ls -l | grep \".c\"\n");

    return 0;
}
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./ls_grep_pipe
```

```
drwxr-xr-x 2 uday uday 4096 Aug 5 04:16 Practical-01
drwxr-xr-x 2 uday uday 4096 Aug 8 03:08 Practical-02
drwxr-xr-x 2 uday uday 4096 Aug 8 02:57 Practical-03
drwxr-xr-x 2 uday uday 4096 Aug 5 03:29 Practical-04
drwxr-xr-x 2 uday uday 4096 Aug 5 03:29 Practical-05
drwxr-xr-x 2 uday uday 4096 Aug 5 03:29 Practical-06
drwxr-xr-x 2 uday uday 4096 Aug 5 03:29 Practical-07
drwxr-xr-x 2 uday uday 4096 Aug 5 03:29 Practical-08
drwxr-xr-x 2 uday uday 4096 Aug 5 03:29 Practical-09
drwxr-xr-x 2 uday uday 4096 Aug 5 03:29 Practical-10
-rwxr-xr-x 1 uday uday 16296 Aug 22 03:20 Practical-4
-rw-r--r-- 1 uday uday 899 Aug 22 03:20 Practical-4.c
-rw-r--r-- 1 uday uday 480 Aug 22 03:26 Zombies_fixed.c
-rw-r--r-- 1 uday uday 1151 Aug 22 04:16 ls_grep_pipe.c
-rwxr-xr-x 1 uday uday 16400 Aug 22 04:11 producer_consumer
-rw-r--r-- 1 uday uday 2489 Aug 22 04:11 producer_consumer.c
```

```
Command executed successfully: ls -l | grep ".c"
```